AE5188 Response to review #1 2026/09/07		

1.1	A more serious objection arises in the n=5 and 6 cases where transformations are not sampled
	exhaustively. The Hermite normal forms are a nice way of enumerating the supercell
	transformations, but they are a bad set of matrices for the given application where they will be
	composed with the 3480 unimodular matrices, because the HNFs often have bigger elements
	than necessary.
	For example, we can take the index-5 supercell pair of 10,10,10,90,90,90 and
	10,22.4,22.4,90,90,90 where b and c are larger by sqrt(5). A simple cell transformation would
	be b'=2b-c, c'=b+2c, thus the transformation we wish to discover is M = [1 0 0 / 0 2 -1 / 0 1 2].
	The corresponding HNF is H = [1 0 0 / 0 5 0 / 0 3 1], which can easily be verified to generate
	the same points as M. By the manuscript's current approach, we write M = UH or M = HU and
	hope that U is one of the 3480 small-entry-unimodular basis transformations. We can easily
	enumerate the 31 HNFs and compute M*H^-1 and H^-1*M to verify that there is no U with the
	required small integer entries. The following Python code reproduces this demonstration:
	import numpy as np
	M = np.array([[1, 0, 0],
	[0, 2, -1],
	[0, 1, 2]])
	hnfs = []
	for a, d, f in [(5, 1, 1), (1, 5, 1), (1, 1, 5)]:
	for b in range(a):
	for c in range(a):
	for e in range(d):
	hnfs.append(np.array([[a, 0, 0],
	[b, d, 0],
	[c, e, f]]))
	assert len(hnfs) == 31
	for H in hnfs:
	U = np.linalg.inv(H) @ M
	integral = np.allclose(U, np.round(U))
	if integral: print(U)
		a bad set of matrices for the given application where they will be composed with the 3480 unimodular matrices,
		
		The reviewer is correct in that the set of matrices is not optimal. The complete set for 5 and 6 is very large (although concurrency might help), so a restricted set has been constructed. If the reviewer can contribute a better, more complete but accessible set, then we would be glad to include it.

1.2	Supercells correspond to sublattices, no? The title "Approximate Supercell (Superlattice) Matching" doesn't sound right to me for that reason.	**Thank you, I removed superlattice from title and the text**
1.3	Table 1: "For n = 3 and n = 4 the complete set is used with the 13 and 35 Hermite normal forms appended, respectively; those matrices are already present in the complete set, so the appended copies are redundant and are retained only as a guard." Why?? It's harmless but strikes the reader as very strange to include these vestigial transformations.	**removed the n=3, n=4 description. The reviewer is correct.**

1.4	Table 1 again: "Left multiplication preserves the lattice and supplies alternate settings of it; right multiplication changes the lattice." The last clause is very mystifying, I made sense of it eventually but I wish it were explained better. "The lattice" here is H*B, a sublattice corresponding to a particular HNF. If we insert the unimodular resetting matrix U, then yes H*U*B is a different lattice. However it was a painful effort to puzzle out that it's different relative to _what_, so I strongly recommend the authors revisit this to try and add some clarity.	**The text has been corrected. What it was supposed to say was that the H and B matrices are multipled in both orders (AB and BA). But that was also not needed and it has been removed.**
1.5	Appendix: I really don't think the code excerpt here is necessary, we can all surmise that it's possible to efficiently compute all the matrices with these entries and this determinant.	**deleted as requrested by the reviewer**

	AE5188 Response to review #2 2026/09/29	
2.1	the comparison of two unit cells. In the present manuscript, the authors report a new feature of this program that allows the relationship between two unit cells to be determined when one is a supercell of the other. The determination of the relationship between two unit cells when one is a supercell of the other has many applications in crystallography. I therefore consider the work presented in this manuscript to be of interest to the crystallographic community. However, the manuscript requires major revision before it can be considered for publication. One of my main concerns is that, in its present form, the methodology and examples are not explained in sufficient detail. Throughout the manuscript, the authors frequently state what the program does or reproduce its output, but provide very limited explanation of the underlying procedure, the meaning of the different quantities and options, and how the results should be interpreted. As a consequence, the manuscript currently reads more like a brief description or documentation of a new program feature than a self-contained scientific article. The manuscript would benefit from a more complete explanation of the methodology, the rationale behind the different steps of the algorithm, and the interpretation and significance of the examples presented. The following comments and suggestions should be addressed in a revised version of the manuscript:	In general, the 2nd reviewer asks for information that is actually in Andrews&Bernstein, 2026b. Repeating that seems unnecessary.
2.2	• The Abstract should be expanded. In its present form, it only states the general aim of the work. It would benefit from briefly describing the methodological development, the validation of the approach, and the scope of the new implementation.	**I do not agree. The abstract should be a brief statement of what the user might expect.**

2.3	I think that the Introduction of the manuscript needs some further work. On page 2, line 018, the authors cite Andrews & Bernstein (2026), but it is not clear to which of the two articles published in *Acta Crystallographica Section A* they are referring: Andrews & Bernstein (2026a) or Andrews & Bernstein (2026b). I would therefore suggest that the authors check this citation and complete the corresponding reference, since the page numbers are missing and the journal title is incomplete.	**the issue with the references has been fixed.**
2.3a	I would also recommend that the authors expand the Introduction and briefly summarize the work presented in their previous papers. As it currently stands, a reader who is not already familiar with these publications may find it difficult to understand the context of the present manuscript and how it builds on the previous work. In particular, the statement ***"A user of that software (CmdLMP3) asked if the same could be applied to the determination of unit cells to supercells. Here we extend their method to the case of supercells"*** (page 2, lines 023-028) could be developed further to explain more clearly why this new implementation is scientifically important and in which crystallographic problems it may be useful.	**I have added a short note about NIST*LATTICE. In general, to a practicing crystallographer, the utility of supercell matching is rather obvious.**
2.4	• I also think that the section describing the algorithm needs further work. On page 4, line 012, the authors state: "The above algorithm suffices for matching lattices that are somewhat similar to each other, but the question of matching a cell to a supercell can be tackled with a modest modification." It is not clear what the authors mean by "the above algorithm": are they referring to the procedure summarized in Figure 1, or to the algorithm described in their previous work? Similarly, it is not clear what exactly is meant by "a modest modification" and which part of the algorithm is being modified. These points should be clarified.	*LCA - add clearer referring to the figure and to the paper.* **The sentence containing "modest" that troubles the reviewer is followed exactly by the explanation that the reviewer requests. To make it clearer, I have changed the sentence with "modest" to end not in a period, but it a colon, followed by the explanation.**
2.4a	In addition, the caption of Figure 1 provides considerably more detailed information about the algorithm than the section specifically devoted to it. The authors should revise this section and provide a clearer and more complete description	Partly, this is just style, but the complete description is in the earlier paper as the

	provide a clearer and more complete description of the algorithm in the main text, rather than leaving essential methodological details mainly in the figure caption.	reviewer cites earlier. **Our preference in this case is to keep the detailed description in the caption, so it is adjacent to the figure.**
2.5	The authors should also discuss the limitations of the search in more detail. Figure 1 states that the $\mathbf{P}_3$ score depends on the setting of the transformed cell, while Table 1 indicates that for n=5 and n=6 the settings of each derivative lattice are sampled less densely. It would therefore be useful to clarify whether this reduced sampling can affect the identification or ranking of the best transformation, under which conditions the best match might be missed, and how the authors have tested the robustness of the procedure.	???? to answer the last question (*how the authors have tested the robustness*), there are examples and the supplementary material that the reviewer cites.
2.6	• In the caption of Fig. 1, the citation to Andrews & Bernstein (2026) should also be clarified. As noted above, it is not clear whether the authors are referring to Andrews & Bernstein (2026a) or Andrews & Bernstein (2026b).	**fixed**
2.6a	The caption of Fig. 1 is also the first place in the manuscript where the terms “reference cell” and “probe cell” are introduced. Since these terms are used repeatedly in the following sections, I would recommend that the authors clearly define them when they are first introduced, or preferably in the main text before Fig. 1. This would make the subsequent description of the algorithm easier to follow.	They are clearly defined in A&B 2026b. **I have added more explanation.**
2.7	The statement on page 4, lines 023-026, referring to supercells “that have a base vector tripled” should be double-checked, as I am not sure that this terminology is correct. A determinant of 3 does not necessarily mean that one basis vector is tripled, rather, it means that the volume of the transformed cell is three times that of the original cell.	**The reviewer is correct, but the statement is explicitly written for the case where a base vector is tripled, not the general case that the reviewer is citing.**
2.8	• On page 4, lines 026 and 027, the statement “Other than that, the algorithm would only need output reporting changed” is unclear. Do the authors mean that, apart from replacing the transformation-matrix library, the matching algorithm remains unchanged and only the format or content of the reported results needs to be modified? This should be explained more clearly.	**That confusing sentence is removed.**
2.8a	• I do not understand the purpose of Section 4 (“Notation”). The section only provides the	**We do not agree with the choice of the**

	definition of $\mathbf{P}_3$, which was already introduced in Andrews & Bernstein (2026a), without explaining why this notation needs to be introduced again here or how it is specifically used in the present implementation. Since $\mathbf{P}_3$ is used as the distance measure for evaluating the different transformations, I would suggest that the authors either integrate this information into the description of the algorithm and explain its role more clearly, or remove this section and refer the reader to the previous publication instead.	**reviewer. It is correct that it is in the referred article, but it is short and informative here, and it is not clear why the reviewer asks for other descriptions from the literature but excludes this. We would like to leave it.**
2.9	• I appreciate that the authors provide in Appendix A the C++ function used to generate the transformation matrices. However, I think that Section 5.1 should explain in more detail how these matrices are generated. At present, the reader has to inspect the source code to understand the procedure. The authors should briefly describe how the integer matrices are constructed, how the range of their elements is selected, and how matrices with the required determinant are identified. This would also help the reader understand the discussion of the number of matrices in Section 5.2.	**On the other hand, reviewer #1 considered the code to be unnecessary. I have removed the code/appendix, and in its place, there is a short algorithmic description in the Algorithm section of the text.**
2.10	• Section 5.2 should also be expanded. At present, the section itself contains only a very brief explanation, whereas the caption of Table 1 provides most of the relevant methodological information. The essential description of how the transformation matrices are generated and selected, particularly the different treatment of indices n=1-4 and n=5-6, should be moved from the caption into the text. The table caption could then be shortened accordingly.	**I have added a short description in the algorithm section. We strongly perfer to have the details in the caption, close to the table which the caption's data describe.**
2.11	• I would also suggest that the authors briefly explain what is meant by the Hermite normal forms (HNFs) introduced in Section 5.2. In particular, it would be helpful to clarify that the HNFs provide representatives of the distinct sublattices of a given index and to explain their role in the generation of the transformation matrices. This would make the discussion of the 7, 13, 35, 31 and 91 derivative lattices in Table 1 easier to follow for readers who are not familiar with this approach. Although explanations of the HNFs are provided in the supplementary material (ae5188sup1.txt), the manuscript would benefit from including a brief explanation in the main text and adding an explicit reference to the relevant part of the supplementary material.	**Excellent suggestion! I have added a very brief note, a reference to Santoro and Mighell, and a reference to the Wikipedia description.**

2.12	• In the caption of Table 1, the authors state that “for n=1 and n=2 the complete set is used in CmdLMP3.” It is not clear what is meant by “the complete set.” From the following explanation, I understand that this refers to the complete set of integer matrices with determinant +n within the selected range of matrix elements [-n,n]. If this is the case, the authors should state this explicitly, since “complete set” could otherwise be interpreted as the complete set of all integer transformation matrices with determinant n.	**There is a column in the table that is labeled "Used in CmdLMP3". How is that not explicit enough?** **and the caption says " the complete set of integer matrices with determinant = 1 or 2, respectively with elements -1,0,1 or -2,-1,0,1,2, respectively is used in CmdLMP3"**
2.13	• Section 5.3 is difficult to follow. It is not clear whether the order in which the two unit cells are provided as input affects the search. Does the program automatically compare their primitive-cell volumes and interchange the reference and probe cells when necessary? If so, this should be stated explicitly.	**I have added an explicit description of the choice.**
2.13a	The authors should also clarify the meaning of “reference” and “probe” throughout this section, since the description in Section 5.3 does not seem fully consistent with the statement “swapped reference and probe cells” reported in Example 2. I would also suggest expanding the explanation of the program output, so that readers can clearly understand the information reported later in the examples in Section 6 and the meaning of the different matrices and quantities presented there.	**DONE**!
2.14	• I would suggest that the authors add an explicit reference in Section 5.3 to the supplementary material (ae5188sup1.txt), where important information is provided on how the program is used, how the input should be introduced, the different search options available, and how the output should be interpreted. At present, much of this information is not explained in the main text. If the authors do not intend to include these details in the text, they should at least refer explicitly to the relevant supplementary material so that readers can easily find this information.	**I do not agree.**
2.15	• In Section 6, the authors provide three examples of the use of the program. However, at present they mainly reproduce the program output without sufficient explanation. I would suggest adding a short discussion to each example explaining why the example was selected, what aspect of the	**I have added some notes about each example.**

	program it is intended to illustrate, and how the output should be interpreted. In particular, it would be helpful to explain what conclusions can be drawn from the reported transformation matrices and matching parameters. The criteria used to classify matches (e.g. “EXCELLENT” or “POOR”), the origin of the 2% threshold, and the meaning and role of the reported $\mathbf{S}_6$ angle should also be explained. In addition, Example 3 appears to contain an incomplete statement (“Validating using quality thresholds: Checking result 1...”), which should be completed or removed.	
2.16	• I think that Section 7 (“Summary”) should be substantially revised. In its present form, the first paragraph mainly states the aim of the manuscript, while the second paragraph only provides information on the computational time required for some determinant values. I do not think that the latter information is necessary in the Summary. More importantly, the Summary should clearly state the main contribution of the work, what has been demonstrated with the new implementation, and the main applicability and limitations of the method. At present, the section does not provide the reader with a clear conclusion from the work presented in the manuscript.	**Perhaps this is a style issue. In my opinion, summary sections are often not useful. In many experimental articles, they can be quite informative, but in a short methods article like this I see little to add that is not already in the abstract and the introduction. I have elected to leave what is there, but I would actually prefer to remove the Summary.**
2.17	• The AI-use statement in the Acknowledgements should be checked, as it still contains the placeholder “[CmdLMP3 module/function names]” (see page 8, line 058). This should be replaced with the corresponding module or function names before publication.	**fixed**